3 June 1971
SUPERSEDING
MIL-P-43477(GL)
30 December 1966

MILITARY SPECIFICATION

PRINTING PRESS, OFFSET, WEB-FED, FOR SPECIAL WARFARE

1. SCOPE

1.1 Scope.- This specification covers 3 types of motor driven, web-fed offset lithographic printing presses capable of printing within a 15-1/2 by 9-7/8 inch image area.

1.2 Classification.- The web-fed, offset printing presses shall be of the following types, as specified (see 6.2):

Type I	-	2-Color Press (1-color one side and 1-color on the opposite side of the web or 2-colors one side of the web).
Type II	-	3-Color Press (1-color one side and 2-colors other side of the web or 3-colors one side of the web).
Type III	-	3-Color Press with U.S. Army press package.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-285	-	Thread, Polyester.
TT-E-529	-	Enamel, Alkyd, Semi-Gloss.
TT-P-636	-	Primer, Coating, Synthetic, Wood and Ferrous Metal.
TT-P-664	-	Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
CCC-C-419	-	Cloth, Duck, Cotton, Unbleached, Plied Yarns, Army and Numbered.
CCC-C-428	-	Cloth, Cotton, Duck, Fire, Water, Weather and Mildew Resistant.
PPP-P-40	-	Packaging and Packing of Hand Tools.

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PPP-T-60 - Tape, Pressure-Sensitive Adhesive, Waterproof, For Packaging.
PPP-B-636 - Box, Fiberboard.

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MIL-C-104 - Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Method Of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture and Fungus-Resistant.
MIL-C-496 - Clips, End, Strap.
MIL-P-514 - Plate, Identification, Instruction and Marking, Blank.
MIL-W-530 - Webbing and Tape, Textile, Cotton, General Purpose.
MIL-B-543 - Buckles, Tongueless and Web Strap.
MIL-C-5015 - Connector, Electric, AN Type.
MIL-M-10578 - Metal Conditioner and Rust Remover (Phosphoric Acid Base).
MIL-F-10884 - Fasteners, Snap.
MIL-V-13811 - Varnish, Waterproofing, Electrical, Ignition.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts, Packaging Of.
MIL-G-16491 - Grommet, Metallic.
MIL-C-43285 - Cloth, Coated (Chloroprene Base Coated, Chlorosulphonated Polyethylene Top Coated).
MIL-P-46093 - Primer Coating, Synthetic (For Brake Drums).

STANDARDS

FEDERAL

FED-STD-595 - Colors.
FED-STD-751 - Stitches, Seams and Stitchings.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirement For Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of.

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DRAWINGS

ARMY NATICK LABORATORIES

6-1-4154 - Printing Press, Offset, Web-Fed, For Special Warfare; Drawing List.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw Thread Standards for Federal Services

U. S. DEPARTMENT OF COMMERCE

PS 1-66 - Soft Plywood/Construction and Industrial.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

NATIONAL FIRE PROTECTION ASSOCIATION

National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Description.- The presses shall conform to the design, details, and dimensions shown on the applicable drawings listed on Drawing 6-1-4154 and as specified herein. The figures included with this specification are furnished for convenience of identification only, and do not establish exact details of design or construction. Each type of printing press shall be capable of continuously printing on a web (continuous paper sheet from a paper roll) at a rate of 25,000 impressions per hour (IPH). The web shall be fed into the press from a roll stand, with brake, through a variable speed mechanism. The press shall have turning bars for turning over the web between the first and second offset printing units, for printing on the reverse side of the web. The delivery unit of each press shall cut the web into 10-1/2 by 16 inch sheets. The sheets shall be collected and delivered to an automatically lowering delivery board. Each press shall be capable of producing a minimum of 160,000 printed sheets within eight hours of operation. Downtime for initial mounting of the plates, inking and adjusting the press, and the initial mounting of the paper roll and webbing of the press shall not be considered part of operating time. For a description of a type I press, see paragraph 3.4.1 and figure 1. For a description of a type II press, see paragraph 3.4.2 and figure 2. For a description of a type III press with U.S. Army press package, see paragraph 3.4.3 and figures 2 and 3.

3.1.1 Major components, printing presses.-

3.1.1.1 Feed roll stand, brake mechanism, and feeder.- The feed roll stand, brake mechanism, and feeder are shown in figure 4. The feed roll stand shall accommodate a 32-inch diameter by 16-inch wide roll of paper. The rolls of paper used shall have a cardboard core having an inside diameter of 3 inches, minimum. The brake mechanism shall maintain drag on the paper roll shaft to provide web tension and braking and stopping of the paper roll without unwinding whenever the press is stopped. The feeder shall pull the web from the roll of paper and feed the web to the first offset printing unit at a regulated speed.

3.1.1.1.1 Feed roll stand.- The feed roll stand shall consist of two support brackets and four roll shaft support bearings.

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3.1.1.1.2 Brake mechanism.- The brake mechanism shall consist essentially of two lateral members housing the brake shoes, (item 10, figure 4); a top member with a brake tension adjusting knob, (item 11, figure 4); a bottom member with the paper roll lateral adjusting knob (item 8, figure 4); and paper roll shaft with brake drum (item 9, figure 4). Expanding chucks shall be provided on the paper roll shaft for mounting rolls of paper.

3.1.1.1.3 Feeder.- The feeder shall consist essentially of a variable speed drive, two feed rollers with rider roller, two tension rollers, a dancing roller, and a dancing roller counter weight with shock absorber.

3.1.1.2 Offset printing units.- All the major parts of an offset printing unit are shown in figures 5, 6, 7 and 8 and identified by the legends on the figures.

3.1.1.2.1 Ink distribution system.- The ink distribution system shall consist of ten rollers, namely; one ink fountain roller, one ductor roller, two Ebonite vibrator rollers, four rubber-covered distributor rollers, and two rubber-covered form rollers. These rollers are shown in figures 5 and 6 and arranged as in figure 7. Ink shall be fed to the ink distribution system from an ink fountain (item 2, figure 5) by means of an ink ductor roller. An adjustable blade and adjusting screws shall regulate distribution of ink on fountain roller. An ink drip pan (item 4, figure 5) shall be provided to catch ink spillage. A movable wash-up attachment (item 12, figure 6) shall be mounted to the side frames to facilitate cleaning of the ink distribution system.

3.1.1.2.2 Water distribution system.- The water distribution system shall consist of five rollers, namely: a water fountain roller, a water ductor roller, a water vibrator roller, and two water form rollers with absorbent covers. These rollers are shown in figures 5 and 6 and arranged as in figure 7. Water shall be fed to this system from the water solution container and holder (item 22, figure 5) which shall control the water level in the water fountain. Water fountain stops (item 21, figure 5) shall be provided to control water flow to the ductor roller from the fountain roller. A water drip pan (item 18, figure 5) shall be provided to catch overflow or spillage.

3.1.1.2.3 Plate cylinder.- The plate cylinder shown in figures 6 and 7, shall be undercut 0.012 inch below bearer surface to accommodate 11-5/8 by 16-3/8 by 0.009 inch metal offset plates. The plate cylinder image surface shall be adjustable to plus or minus 1/8 inch from side to side to provide for side registration. The cylinder shall have a locking device for clamping the metal offset plate to the cylinder.

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3.1.1.2.4 Blanket cylinder.- The blanket cylinder, shown in figures 6 and 7, shall be undercut 0.071 inch below bearer surface to accommodate a 11-9/16 by 16-5/8 by 0.065 inch 3-ply blanket. The blanket cylinder shall have a device for clamping and tightening the blanket to the cylinder.

3.1.1.2.5 Impression cylinder.- The impression cylinder, shown in figures 6 and 7, shall be a smooth surfaced cylinder that forces the web in contact with the blanket cylinder. The impression cylinder shall have an adjustment to permit changing the distance between the impression and blanket cylinders for accommodating paper varying from 9-pound manifold to 90-pound index.

3.1.1.3 Delivery unit.- The delivery unit shall be as shown in figures 9, 10 and 11 with major parts identified by the legends on the figures. The delivery unit rotary knife assembly shall cut the printed web into cut sheets at 10-1/2-inch intervals. The cut sheets shall be delivered by a conveyor assembly to an automatically lowering delivery table with casters (item 10, figure 10). An intermediate support delivery board shall be provided to receive the sheets at a reduced speed while the printed pile is being removed and an empty delivery table is placed in operating position.

3.1.1.4 Operating controls.-

3.1.1.4.1 Offset printing units and main control station.- Operating control stations shall be provided at each offset printing unit (see figure 8) on the operator's side of the press to permit forward or reverse jogging and stopping of the offset printing unit. Starting, stopping, and speed control of the press with air blower and impression controls shall be handled from the main control station (see figure 11) on the operators's side. The legends on figures 8 and 11 identify all the operating controls.

3.1.1.4.2 Variable speed drive control.- The variable speed drive control (item 5, figure 4) shall be operated from the operator's side and shall permit sychronizing the speed of the web feed with the speed of the offset printing units.

3.1.1.4.3 Register controls.- Register control (item 1, figure 11) shall provide for register of the printed image between offset printing unit 1 and 2. Register control (item 1, figure 11) shall provide for register of printed images between offset printing units 2 and 3 (on 3-color presses only). Cut-off register control (item 10, figure 11) shall provide for register between the printed image and cut-off.

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3.1.1.4.4 Automatic controls.- The press shall be equipped with automatic controls to stop the press whenever there is: (1) a web breakage, (2) a jam prior to the cutting knife, and (3) when the paper has reached a minimum thickness of 1/4 inch on the paper roll core. As a safety feature, shear pins shall be located between the cutting knife cylinder and drive gear for prevention of damage in the event of a severe pile-up. A mechanical tripping device shall be provided to stop the press during a jam prior to cutting knife.

3.1.1.4.5 Miscellaneous controls.- A delivery clutch operated by a handle (item 21, figure 11) shall be located between the last offset printing unit and the delivery unit to disengage the delivery unit from the offset printing units. Manually operated electric switches located on the main control station (item 22, figure 11) shall be provided to throw the blanket cylinders "ON" or "OFF" impression (items 17, 18, 19 and 20, figure 11). Additional manually operated handles and knobs shall be provided for controlling various functions of the offset printing units, such as water-form roller (item 12, figure 8) and ink-form roller control, (item 4, figure 8) impression control, (item 11, figure 8) and side register control (item 5, figure 8) (see 4.5.2). A manually operated control knob (item 8, figure 4) shall be provided for laterally adjusting the paper roll in the feed roll stand (see 4.5.1.1).

3.1.1.5 Auxiliary equipment.-

3.1.1.5.1 Blower and motor unit.- A blower (item 15, figure 3) shall be provided on all types of presses to supply an air cushion between delivered leaflets. The blower shall be V-belt-driven by means of a 1/2 horsepower, 208 volt, 60 Hertz (Hz), 3 phase, AC motor.

3.1.1.5.2 Variable speed drive unit.- A variable speed drive unit (item 49, figure 2) shall be provided on all types of presses and shall have a 30 to 1 speed ratio with vernier control for regulating the speed of the rollers.

3.1.1.5.3 Speed indicator.- The speed indicator (item 6, figure 11) shall have a 0-40,000 scale to indicate the press speed in impressions per hour.

3.1.1.5.4 Sheet counter.- The sheet counter (item 4, figure 11) shall have a six-digit counter. The sheet counter shall be manually set to zero, and shall be capable of being engaged or disengaged while the press is running.

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3.1.1.5.5 Automatic lubricator.- An automatic lubricator shall provide a supply of oil for the pressure lubrication system (item 17 figure 3) on all types of presses. The lubricator shall contain a 3-pint oil reservoir with sight level indicator. A manually operated push button shall provide for bypassing the automatic cycling control.

3.1.1.5.6 Press leveling devices.- A two-bubble level shall be located on the base of the frame between the last offset printing unit and the delivery unit. To level the press, one bubble level shall be lined up with the long axis of the press and the other with the short axis.

3.1.1.5.7 Static eliminators.- A static eliminator system shall be provided to eliminate static electricity generated by the paper. The system shall consist of static eliminator bars, transformer, indicator light and necessary wiring and fittings. The exact location of eliminator bars to be determined while operating the press with a web.

3.1.1.5.8 Lifting bars.- Lifting bars shall be provided to lift the press by means of an overhead hoist. These bars shall be easily removed for storage when they are not required for lifting purposes.

3.1.1.5.9 Motor generator unit.- The motor generator unit (item 21, figure 3) shall be required for all types of presses to supply the direct current power for operation of the press. This unit shall be suitable for operation on the power specified in 3 5.8.1.

3.1.1.5.10 Main drive motor.- A 7-1/2 horsepower main drive direct current motor (item 16, figure 3) controlled from the main control station shall supply power for all types of presses. The main drive motor shall receive its power supply from the motor generator unit.

3.1.2 Supporting equipment.- Supporting equipment required shall be as follows:

3.1.2.1 Plate bending jig.- The plate bending jig is shown on figure 12, and shall be used for bending plates for the plate cylinder.

3.1.2.2 Type III, 3-color press with U. S. Army press package.- The components specified in 3.1.2.2.1 through 3.1.2.2.9 are applicable to the U. S. Army press package part of the type III, 3-color press with U. S. Army press package (see 3.4.3).

3.1.2.2.1 Ramp plates.- Ramp plates (items 2 and 4, figure 3) shall be of plywood construction with skid-proof faced steel plates for use in loading and unloading of material and supplies to and from the platform base.

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3.1.2.2.2 Platform extension plates.- Platform extension plates (item 1 and 3, figure 3) shall be made of plywood with skid-proof faced steel plates and equipped with six adjustable legs. The platform extension plates shall be used to extend the platform base on the operators side of the press.

3.1.2.2.3 Platform base ramp.- The platform base ramp (item 25, figure 3) shall be made of steel and used to allow easy removal of printed leaflet piles.

3.1.2.2.4 Tables.- The four tables (items 5 and 7, figure 3) shall be of the folding leg type for use as work tables during press operation.

3.1.2.2.5 Storage cabinets.- The storage cabinets (items 22, 23, 24, 26, 28, and 30, figure 3) shall be made of steel. Each cabinet shall be provided with two eye-bolts on top to facilitate the lifting of the cabinets for operational set-up.

3.1.2.2.6 Ink mixing slabs.- The three ink mixing slabs (item 27, figure 3) shall be demountable type, with corrosion resisting steel faces bonded to plywood core.

3.1.2.2.7 Platform base.- The platform base (item 13, figure 3) shall be a welded steel unit used as an operating base for the printing press and as a shipping platform for all the components of the type III - 3 color press with U. S. Army press package.

3.1.2.2.8 Shipping frame.- The shipping frame (item 6, figure 3) shall be of the knock-down type fabricated from steel structural members, for use as a protective frame and support for cabinets, tables and extension platforms during shipment.

3.1.2.2.9 Covers.- For fabric covers see 3.10.

3.1.2.2.10 Installation tools.- For installation tools see 3.6.1.

3.2 Alternate components.- Components offered as equivalent to components specified hereinafter and on the referenced drawings and bills of materials, as a specific manufacturer's part number or equal shall be functionally equal to the manufacturer's part number identified and of equal or better quality. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components, unless such

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modification or change is specifcally approved by the contracting officer. Prior to manufacture of the initial production unit, the supplier shall submit for the contracting officer's approval a list identifying each propose alternate component, together with proof that each listed component complies with requirements specified above. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of production and does not relieve the supplier of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.3 Materials.- Materials shall conform to the applicable specifications, standards, and requirements specified herein and on the applicable drawings. Materials not specifically defined shall be material regularly used for the purpose by the manufacturer.

3.3.1 Varnish.- The fungus resistant varnish shall be in accordance with MIL-V-173.

3.3.2 Threaded fasteners, nuts, washers and set screws.- All threaded fasteners, nuts, washers, and set screws shall be made from series 300 corrosion-resisting steel, or cadmium plated carbon or alloy steel.

3.3.3 Finish materials.-

3.3.3.1 Metal conditioner.- Metal conditioner shall be type I or II conforming to MIL-M-10578.

3.3.3.2 Primer, metal.- Metal primer shall conform to TT-P-636.

3.3.3.3 Enamel.- Enamel shall conform to TT-E-529 in the color number shown in table I and shall approximate the corresponding color number in FED-STD-595.

TABLE I.- Enamel colors

Color No.	Application
Red 21136	Lubrication points
Gray 26134	Printing press
Green 24533	Cabinet interior surfaces
Olive drab X24087	Cabinet exterior surfaces
Olive drab X24087	All other components

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3.3.4 Plywood.- Plywood shall be marine exterior type conforming to PS 1-66. Thickness shall be as indicated on the applicable drawings or 3/4 inch when not specified on the drawings. Face surfaces exposed to view in use shall be A grade.

3.4 Design.-

3.4.1 Type I, 2-color press.- The type I, 2-color press (figure 1) shall be designed to print 1 color on one side of the web and one color on the opposite side or 2 colors on one side of the web.

3.4.2 Type II, 3-color press.- The type II, 3-color press (figure 2) shall be designed to print 1 color on one side of the web and 2 colors on the opposite side or 3 colors on one side of the web.

3.4.3 Type III, 3-color press with U. S. Army press package.- The type III, 3-color press (figure 2 and 3) with U. S. Army press package shall consist of a type II, 3-color press with the addition of supporting equipment specified in 3.1.2.2 through 3.1.2.2.9.

3.5 Construction.- Construction of type I, II and III presses shall be as hereinafter specified and as shown on the applicable drawings. Details not specifically covered shall be in accordance with the manufacturer's regular commercial practice.

3.5.1 Castings and forgings.- All castings and forgings, when applicable, shall be free from visible blowholes, porosity, hard spots, shrinkage defects and cracks in the finished items. Forgings shall be free from visible scale, inclosures, cold shuts, mismatchings and other defects that may affect the structural strength of the finished forgings involved. Chemical composition of the castings and forgings shall be adequate to meet the performance requirements specified herein without failure.

3.5.2 Machined components.- Machined components shall be within the dimensional tolerances established on the drawings. Surface finish shall not adversely affect the proper alignment or functioning of mating parts. All rough and sharp edges shall be removed. Threaded portions shall conform to Handbook H28, for the size thread required. The leading end of all threaded portions subject to assembly and disassembly shall be chamfered a distance at least equal to the pitch and depth of the thread.

3.5.3 Sheet metal components.- Unless otherwise specified on the applicable drawings, all sheet metal components shall be made from not less than 16 gage (.0598) steel sheets or strips. Sheet metal components shall be within the dimensional tolerances established on the drawings. All like bends shall be of uniform radius and unless otherwise specified

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on the drawings, radii shall be as small as practicable in relation to sheet metal thickness. Bends of a major character, and embossing, when applicable, shall be accomplished using metal dies and fixtures to insure uniformity of size and shape. All exposed sheet metal edges shall be dressed to remove sharp edges and burrs.

3.5.4 Riveting.- Rivet holes shall be drilled or punched to insure a snug fit with the rivet shank after heading. When practicable, rivet holes shall be drilled or punched at time of assembly. Burrs at rivet holes that may prevent proper seating of rivet heads, shall be removed. Heading shall be in full contact with mating surfaces, concentric with the rivet shank, and rivet heads shall be free from splits, burrs, and sharp edges after heading.

3.5.5 Welding.- The surface of parts to be welded shall be clean and free from scale, paint, grease and foreign matter. When aluminum, corrosion-resisting steel or other ferrous metal alloy is welded, a welding rod of a chemical composition compatible with the parts being welded shall be used. Welds shall be uniformly run without voids and undercuts. All welding scale shall be removed after welding has been accomplished, and all projections and rough edges shall be dressed smooth. Spot welds, when utilized, shall be sound, smooth and free from burn through, pits and flashes.

3.5.6 Threaded connections.- Unless otherwise specified on the drawings, threaded connections shall not be peened, staked or otherwise deformed to prevent disassembly of the threaded connections.

3.5.6.1 Pipe thread connections.- A commercially available, non-hardening anti-sieze thread compound shall be applied to pipe threads prior to assembly in the line.

3.5.7 Metallic pipe, tube and conduit bending.- All pipe and tube bends, when required, shall be of uniform radius and free of kinks, buckles, and noticeable deformation in cross section. All electrical conduit bends shall be in accordance with the applicable provisions of the National Electrical Code.

3.5.8 Electrical.-

3.5.8.1 Power source characteristics.- The type I, II and III presses, and associated equipment, shall be suitable for operation on 120/208 volt, 3 phase, 4 wire, 60 Hz, AC, power source.

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3.5.8.2 Electrical wiring.- All electrical equipment shall be furnished fully wired to terminal boxes with marked terminals and correspondingly marked interconnecting cables or wiring assemblies where necessary. Unless otherwise shown on the applicable drawings, wiring shall be accomplished in accordance with applicable provisions of the National Electrical Code.

3.5.8.3 Fungus and moisture resistance.- The electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to 3.3.1 with the following exceptions:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 0.001 inch to component or element surfaces previously cleaned and prepared.

3.5.8.4 Electromagnetic compatibility.- The end item shall conform to the class IIIB requirements of MIL-STD-461 (see 4.5.3).

3.5.9 Lubrication.- All printing press friction points and bearings, subject to overheating, with resulting excessive wear, shall be lubricated from a common lubrication system piped from the pressure type oil reservoir, insofar as is practicable. All other friction points and bearings shall be adapted for manual application of lubricants. All component parts of the automatic lubrication system shall be of one manufacturer's design.

3.5.10 Textile components (when applicable).-

3.5.10.1 Thread breaks.- Thread breaks in stitching shall be overstitched not less than 1-inch at each break.

3.5.10.2 Stitching ends.- Stitching ends shall be overstitched not less than 1-inch except where ends are turned under in a hem, are held down by other stitching or where stitching is performed automatically. When stitch patterns such as box, box with cross-stitch, "W" stitching or straight line tacking are performed automatically, at least 3 or more tying, over lapping or backstitches shall be used to secure the ends of stitching.

3.5.10.3 Thread lubricant.- The addition of any lubricant to the polyester sewing thread, prior to, or during the sewing operation is prohibited.

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3.5.10.4 Setting of grommets and snap fasteners.- Holes to receive grommets and snap fasteners shall be smaller than the O.D. of the grommet barrel or fastener tube, so that the parts must be forced through the holes. Grommets and fasteners shall be securely clinched without cutting the materials.

3.5.10.5 Setting of end clips.- The webbing shall be inserted full depth of the clip. Clips shall be securely attached without cutting the webbing and the webbing shall be flat at the point of entrance.

3.5.10.6 Seam leakage.- There shall be no dripping of water through the fabric covers, or visible wetting along the seams when the covers are tested as specified in 4.5.5.

3.6 Tools.-

3.6.1 Installation tools, type III, 3-color press with U. S. Army press package.- Installation tools, listed in table II, shall be furnished with each type III, 3-color press with U. S. Army press package.

TABLE II.- Installation tools

Item No.	Description	Quantity
1	Roller skid (fixed). Stokvis Multiton Corp. Mark 4, or equal, modified per applicable drawing	2
2	Roller skid (swivel) with connector F. Stokvis Multiton Corp. Mark 4 S, or equal, modified per applicable drawing	1
3	Ratchet lowering Jack. Simplex Mfg. Co. No. 84A, or equal	2
4	Handle for lowering Jack. Simplex Mfg. Co. No. H84A-H, or equal	2

3.6.2 Maintenance tools.- Maintenance tools, for presses, shall be furnished with each type of printing press, in the quantity and variety specified (see 6.2).

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3.7 Operating supplies, type III - 3-color press with U. S. Army press package.- Operating supplies shall be furnished with each type III - 3-color press with U. S. Army press package in the quantity and variety specified (see 6.2).

3.8 Electric cable, type III, 3-color press with U. S. Army press package.-

3.8.1 Interconnecting cables.- SO type interconnecting cables with waterproof-disconnect connectors shall be furnished. The cables shall consist of one 17-foot cable for blower motor connection; one 17-foot cable for static eliminator connection; one 17-foot cable for main drive motor; one 17-foot cable for press controls; and a 50-foot main power cable.

3.8.2 Lead-in cable.- Not less than one hundred feet of SO type power cable, with waterproof disconnect adapters, to permit wiring the motor generator to the main power source, shall be furnished.

3.8.3 Lighting cables.- Six lighting cables shall be provided to interconnect storage cabinets to power supply box. Four type SO cables, 10 feet long, with grounding type plug connectors shall be furnished to interconnect cabinets. Two type SO cables 25 feet long with grounding type plug connectors shall be furnished to tie in the last two storage cabinets and to connect to the power supply box.

3.9 Spare parts.- Such spare parts as are specified (see 6.2) shall be furnished with each printing press, and shall include one additional offset blanket assembly for each offset unit of the press (see item 3-33, Drawing 6-1-4036) and one additional set of ink rollers and water rollers for each offset unit.

3.10 Covers, fabric, type III, 3-color press with U.S. Army press package. Fabric covers for the motor generator, press, and packaged press shall be as shown on applicable drawings.

3.11 Finish.-

3.11.1 Wood surfaces.- All exposed wood surfaces except those treated with a wood preservative in accordance with applicable drawings, shall be sanded smooth, cleaned of all foreign material and given two (2) coats of clear lacquer or varnish, applied in accordance with the manufacturer's standard practice. The finished surface shall be smooth, completely coated and free from runs, sags and teardrops.

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3.11.2 Metal surfaces.- All uncoated metal surfaces regularly painted by the manufacturer in his commercial practice shall be prepared, primed and enameled as specified in 3.11.2.1 through 3.11.2.3 and shall include all interior and exterior surfaces of cabinets, work benches, printing press platform (type III press) and like items, as applicable. Previously painted surfaces shall be enameled a color to match the color of the component to which attached except when the characteristic color associated with the component may be obscured.

3.11.2.1 Surface preparation.- Uncoated metal surfaces requiring a final enamel finish shall be prepared for painting by application of either type metal conditioner specified in 3.3.3.1. Application shall be in accordance with the metal conditioner manufacturer's recommendations.

3.11.2.2 Priming.- Surfaces prepared as above shall be given a coat of primer specified in 3.3.3.2. The primed surfaces shall be uniformly and completely coated and thoroughly dried prior to enameling.

3.11.2.3 Enameling.- Surfaces requiring a final enamel finish shall be coated with enamel specified in 3.3.3.3 in the color noted therein. The enamel finish shall level out to produce a smooth, uniform surface free from runs, wrinkles, grit and areas of no film.

3.12 Marking for identification.- End item marking for identification shall conform to the unit marking requirements of MIL-STD-130, appearing on a plate conforming to MIL-P-514.

3.12.1 Lubrication points.- For convenience of identification, all points requiring manual lubrication shall be identified by means of a 1/2-inch diameter red painted circle surrounding the lubrication point. Enamel shall be in accordance with 3.3.3.3.

3.12.2 Electric motors.- Electric motors shall have a securely attached corrosion-resistant metal nameplate containing the manufacturer's name, model number, phase, Hz, voltage, rated horsepower, and allowable temperature rise.

3.12.3 Electrical controls.- All manually operated electrical controls shall have a permanently attached identification plate located adjacent to the control. In addition, any necessary warnings should be included on the identification plate to prevent misuse of the control.

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3.12.4 Mechanical controls.- Manually operated mechanical controls shall be identified as in 3.12.3.

3.13 Performance.-

3.13.1 Presses.- Each type I, II, and III press furnished shall be set up and tested as specified in 4.5.1 through 4.5.2.5, and shall produce printed copies as specified therein. During test runs, the amperage reading of electric motors shall not exceed the nameplate rating (see 4.5.2.4). In addition, bearings, shafts, gears and like friction points shall not be heated beyond handling temperature (see 4.5.2.5).

3.13.1.1 Speed setting control.- The control that regulates the IPH speed (see 3.1.1.4.2), shall be such that the press cannot be started from a dead stop until the dial has been first set to zero and then gradually advanced to the desired IPH speed (see 4.5.1.2).

3.13.1.2 Automatic press shut down.- The setting of the paper roll limit switch (see 3.1.1.4.4) and the arm bearing against the paper roll in the feed stand, shall be adjusted to stop the press when the paper roll has been consumed to a minimum of 1/4-inch of the outside diameter of the paper roll core. Each web break stop (see 3.1.1.4.4), bearing against the web during press operation, shall be set to stop the press, when the web breaks and the stop falls below the normal web level. In addition, the mechanical tripping device at the delivery section (see 3.1.1.4.4) shall shut down the press whenever the constant delivery of cut sheets has been interrupted. Tests for automatic shut down shall be conducted in accordance with 4.5.1.2.

3.13.1.3 Acceleration and deceleration.- The presses shall accelerate from dead stop to maximum speed within a maximum of 30 seconds, and decelerate from maximum speed to dead stop within a maximum of 2 seconds (see 4.5.1.3).

3.13.1.4 Printing quality.- Each offset unit shall consistently produce a solid colored image over the entire image area (9-7/8 by 15-1/2 inches) of each sheet printed during the test run in 4.5.2.1. In addition, when tested in accordance with 4.5.2.2 and 4.5.2.3, there shall be no local shaded areas over solid backgrounds, and half tone areas examined under 9-power magnification shall show no appreciable increase in dot size or filling in of shadow detail, when compared with the original press plate detail. Also, the register marks of each offset printing unit shall be exactly overlaid without evidence of multiple images.

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3.13.2 Cabinets and tables, type III U. S. Army press package.- When tested as specified in 4.5.4, all cabinet doors, drawers, and other moving parts shall be operable without binding. All doors and drawer faces shall be parallel with the cabinet faces or frame and all cabinet and door catches shall engage as intended.

3.14 Workmanship.- The end item shall be free of defects such as loose, bent, broken or missing components, rough or sharp edges, and misaligned or improperly assembled components. When applicable, fabric components shall be clean and free of holes, cuts, tears, or cloth defects such as smashes, weak places, or broken or missing yarns. Webbing shall not be frayed or scalloped edged. There shall be no visible evidence of run-offs, twists, pleats, puckers, open seams, or needle chews. All thread ends shall be trimmed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection of components and materials.- In accordance with 4.1, the supplier is responsible for insuring that components and materials were manufactured, tested, and inspected in accordance with the referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.3 Intermediate inspection.-

4.3.1 Intermediate visual and dimensional examination.- Components or sub-assemblies which cannot be checked or measured after final assembly shall be examined at various stages of manufacture for compliance with this specification and applicable drawings. The examination shall be made at an appropriate stage for defects in design, construction, workmanship, dimensions, material and finish. Any deviation from specified requirements shall cause rejection of the component or sub-assembly until corrections have been made.

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4.3.2 Process examination.- Examination shall be made at the appropriate stages of manufacture to determine that applicable operations and processes are accomplished in accordance with specified requirements. Whenever a deviation is noted, correction shall be made to the operation and process, as applicable. Failure to make correction shall be cause for rejection of the affected end item.

4.3.3 Certification of compliance.- Components and materials listed below, may be accepted on the basis of the contractor's certificate of compliance for characteristics indicated.

Component or material	Characteristic	Requirement paragraph
Electrical	Fungus resistance	3.5.8.3

4.4 End item inspection.-

4.4.1 Visual examination.- Each fully assembled end product shall be examined for completeness of assembly; presence and adequacy of components, equipment, accessories, as applicable, and finish, marking and quality or workmanship, to assure the end product complies with specified requirements and drawings. Any deviations noted shall cause rejection of the end product until correction has been made.

4.4.2 Dimensional examination.- Examination shall be made of each complete printing press, and the equipment related thereto, for defects in dimensions. Any dimension not within the specified tolerance shall be corrected prior to acceptance.

4.4.3 Testing of the end product.- Each unit shall be tested in accordance with 4.5.1 through 4.5.2.2, 4.5.2.3 through 4.5.2.5 and, when applicable, 4.5.4 thru 4.5.5.1. The tests specified in 4.5.2.2.1 and 4.5.3 shall be performed on the initial production unit.

4.4.4 Examination of preparation for delivery.- An examination shall be made to determine that preservation, packaging, packing and marking as required by section 5 of this specification are complied with. Whenever a deviation is noted, correction shall be made. Failure to make correction shall be cause for rejection of the affected end item until corrections have been made.

4.5 Tests.-

4.5.1 Performance tests.- The supplier shall furnish all necessary equipment, supplies, personnel, facilities, material and incidental services necessary to perform tests on each completely assembled printing press, including auxiliary equipment and applicable testing equipment. The text of material to be printed and the color of impressions shall be as specified (see 6.2).

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4.5.1.1 Preparation for testing.- Each printing press, including the applicable auxiliary equipment and associated equipment shall be set up, prepared for printing, and operated. The adjustment of the paper roll on the feed roll stand shall be checked for lateral operation (see 3.1.1.4.5). Any errors or omissions noted during set-up, preparation, and test runs shall be recorded for correction.

4.5.1.1.1 Paper, printing.- Paper used for the printing shall be 50-pound basis offset white book paper in roll form, 32 inches in diameter, 16 inches wide and 3 inches inside core diameter.

4.5.1.2 Safety stops.- The speed control setting (see 3.13.1.1) and automatic press shutdown (see 3.13.1.2) shall be tested without paper in the press, ink in the reservoir or water in the fountain. The speed control dial shall be set at zero IPH. The paper roll limit switch arm, and each web break stop shall be tied up with twine to the height assumed during normal press operation, and the mechanical tripping device at the delivery unit shall be set to prevent tripping. With the power supply "on", the speed control setting dial shall be advanced to a press speed of approximately 25,000 IPH (see 3.1.1.5.3). With the press running at this speed the mechanical tripping device at the delivery unit shall be tripped. This action should cause the press to stop. Any nonconformance shall constitute failure to pass test. Noncompliance with the requirement in 3.13.1.1 shall be evident if the press fails to stop or runs prior to first resetting the speed control dial to zero prior to advancing the dial to any desired number of IPH. The feed roll stop and each web break stop shall be tested by cutting the twine with scissors, permitting the stop arm to fall below running position. The action of each stop shall be tested separately. Failure to comply with 3.13.1.1 and 3.13.1.2, as applicable, shall be evident if the press fails to stop when the twine is severed, or if the press will run prior to elevating all stops to running position or resetting the speed setting control dial to zero before advancing to the desired number of IPH. Any nonconformance shall constitute failure to pass the test.

4.5.1.3 Acceleration and deceleration.- After allowing each press to run a sufficient length of time to make the proper initial adjustments as required, the webbed press shall be stopped and then accelerated to 25,000 IPH and back to dead stop not less than six times, to determine the ability of the press to accelerate to 25,000 IPH within a maximum of 30 seconds, and to decelerate to a dead stop within a maximum of 2 seconds (see 3.13.1.3). Any nonconformance shall constitute failure to pass test.

4.5.2 Test runs.- All test runs shall be actual printing operations with all press components functioning as intended including miscellaneous controls (see 3.1.1.4.5) and all offset printing units printing with different color printing inks.

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4.5.2.1 Ink coverage.- With 50-pound book paper in the press, each offset unit shall be individually tested by printing a solid color image over the entire image area (9-7/8 by 15-1/2 inches) of each sheet subsequently cut to 10-1/2 by 16 inch size. A minimum of 2,000 sheets shall be printed by each offset printing unit within a maximum five minute period, for evaluation of ink coverage for compliance with 3.13.1.4. Any nonconformance shall constitute failure to pass the test.

4.5.2.2 Normal test run printing on both sides of the web.- Each press shall be operated in a normal manner up to 8 consecutive hours. Time to change paper roll shall be included within the 8-hour run. During this time a minimum of 160,000 sheets shall be printed on both sides and cut to 10-1/2 by 16-inch size. If 160,000 printed sheets are printed in less than 8 hours, the test shall be terminated and considered as satisfactory evidence of meeting the minimum 160,000 printed sheets within the 8 hour operation requirements (see 3.1). Approximately 500 of the printed sheets shall be furnished the Government for retention and approval of printing quality, ink density and coverage, registration accuracy, consistency of sheet size, (see 3.13.1.4). The 500 samples shall consist of the first 250 and last 250 copies printed during the 8 hour production run. The remaining sheets shall be the property of the manufacturer. Any nonconformance shall constitute failure to pass the test.

4.5.2.2.1 Initial production unit endurance test. Using the same procedure specified in 4.5.2.2, the initial production unit shall be tested with varying speeds up to a maximum of 25,000 IPH for 2 hours during 4-hour periods. The press shall be shut down for a time not to exceed 30 minutes at the end of each 4-hour period to check for excess heating of bearings and electrical and mechanical components. This test shall be conducted continuously for a period of 72 hours including the shut down time. Any nonconformance shall constitute failure to pass

4.5.2.3 Normal test run printing on one side of the web.- Each press shall be operated in a normal manner up to 2 consecutive hours. Time to change the paper roll shall be included within the 2-hour run. During this time a minimum of 40,000 sheets shall be printed on one side and cut to 10-1/2 by 16 inch size. Approximately 500 of the printed sheets shall be furnished the Government representative for retention and approval of printing quality, ink density and coverage, registration accuracy and consistency of sheet size (see 3.13.1.4). The 500 samples shall consist of the first 250 and last 250 copies printed. The remaining sheets shall be the property of the manufacturer. For this test, if 40,000 sheets are printed in less time than the 2-hour production run period, the test shall be terminated and considered as satisfactory evidence that the minimum 160,000 printed sheets can be satisfied. Any nonconformance shall constitute failure to pass the test.

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4.5.2.4 Amperage loads.- The amperage load of electric motors shall be checked for compliance with 3.13.1 during the test run in 4.5.2.2 and 4.5.2.3. An ammeter shall be properly connected to each electric motor and observed at frequent intervals during the start, running, and slow down cycles throughout the production test. Any nonconformance shall constitute failure to pass the test.

4.5.2.5 Bearing temperature.- Bearing temperature shall be checked for compliance with 3.13.1 during the test run in 4.5.2.2. At frequent intervals throughout the test run, non-hazardous friction points shall be touched with the bare hand. Excessive heating shall be evident if the bare hand cannot remain on the bearing surface or housing. Hazardous areas shall be similarly tested each time the press has come to a complete stop. Any nonconformance shall constitute failure to pass the test.

4.5.3 Electromagnetic compatability.- When electromagnetic compatibility is required, the preproduction sample or initial unit of production, as applicable, shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test report and the requirements meet MIL-STD-461.

4.5.4 Tables and cabinets.- When applicable, all cabinet doors and drawers shall be operated several times and examined to determine compliance with 3.13.2. Any nonconformance shall constitute failure to pass the test.

4.5.5 Seam leakage test.- When fabric covers are applicable (see 3.10) the covers shall be tested for leaks in three areas as follows: (1) on top panel seams (2) on seams joining top to wall and (3) on wall panel seams.

4.5.5.1 Procedure.- Suspend the section of the cover containing the area to be tested between two supports (boxes, wooden horses or other suitable support) so as to allow both sides of the fabric to be viewed. The area to be tested shall then be cupped into a hoop (embroidery hoop) 6 to 7 inches in diameter, with the seam running through the center of the cupped fabric. Pour into the cupped area one pint of cold tap water. Allow to stand approximately 5 minutes. At the end of this period, examine the test area for water dripping through the seam or visible wetting along the seam (see 3.5.10.6). Any nonconformance shall constitute failure to pass the test.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Preservation.- Preservation shall be either as specified in Option I or Option II at the discretion of the supplier. Preservation shall be in accordance with the method and material requirements of MIL-P-116 as hereinafter specified.

5.1.1.1.1 Option I.-

5.1.1.1.1.1 Threaded and unpainted exterior surfaces.- Threaded and unpainted exterior ferrous metal surfaces of items such as springs, stands, bases, pins, linkages, grease fittings, bolts and nuts, or any other unpainted metal surface not specifically identified herein, requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. Unpainted ferrous metal information plates shall be coated with varnish conforming to MIL-V-13811.

5.1.1.1.1.2 Exposed machined metal surfaces.- Exposed machined ferrous metal surfaces such as shafts, worm gears, levers, rollers, roller ends, cylinders and working parts of the equipment shall be coated with type P-6 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2. The wrap or cover shall be secured in place with tape conforming to PPP-T-60, type and class optional.

5.1.1.1.1.3 Enclosed gears.- Gears operating in engine lubricating oil (grade 10, 30, or 50) shall have the housing drained and filled to the operating level with type P-10, grade 1, 2 or 3 preservative, as applicable, and tagged to indicate the preservative oil is suitable for operation and shall remain in the housing until the first required lubricant change. Gears operating in other than engine lubricating oil shall have the housing filled to the operating level with the approved lubricant required by the lubrication order for operation and tagged to indicate the housing is filled with the required operating oil.

5.1.1.1.1.4 Exposed chains.- Exposed drive chains shall be coated with enough type P-3 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins and bushings. After the excess preservative has drained, the chains (except lithographic press chains) shall be coated with type P-1 preservative. Chains, if removed, shall be preserved as specified herein and the chains shall be individually coiled. Each coil shall be secured with soft annealed wire and each coil shall be wrapped with barrier-material conforming to MIL-B-121, type I, grade A, class 2. The wraps shall be secured with tape conforming to PPP-T-60, type and class optional.

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5.1.1.1.1.5 Drive belts and pulleys.- Drive belts shall be removed or released from tension. Pulley faces or grooves shall be coated with primer conforming to MIL-P-46093. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1.

5.1.1.1.1.6 Electric motors.- Electric motors shall be preserved in accordance with MIL-E-16298, level A, using the alternate method specified therein for assembled machines.

5.1.1.1.1.7 Controls and wiring.- Exposed ends of wires, electrical cables, plug openings, terminals and openings into switches, junction boxes, and couplings shall be covered or sealed with tape conforming to PPP-T-60, type and class optional.

5.1.1.1.1.8 Spare parts.- The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve and package the spare parts.

5.1.1.1.1.9 Maintenance and installation tools.- Maintenance and installation tools shall be preserved in accordance with the level A requirements of PPP-P-40.

5.1.1.1.1.10 Technical publications.- Technical publications shall be preserved method IC-1 of MIL-P-116.

5.1.1.1.2 Option II.- Each press shall be preserved method IIA. Fabric cover shall be installed over the type III press prior to inclosing within barrier material.

5.1.1.2 Packaging.-

5.1.1.2.1 Type I and II printing presses.-

5.1.1.2.1.1 Technical publications.- Technical publications not preserved method IC-1 shall be placed in a paper envelope and secured to the press in a prominent position.

5.1.1.2.1.2 Plate bending jig.- Each plate bending jig shall be packed in a snug-fitting fiberboard box conforming to type CF (variety SW) or SF, class domestic, style RSC of PPP-B-636. Fiberboard inserts or other cushioning material shall be used to immobilize the contents. Each box shall be securely closed in accordance with the appendix of PPP-B-636.

5.1.1.2.1.3 Eyelet machine.- Each eyelet machine, together with the associated punches, shall be packaged as specified in 5.1.1.2.1.2.

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5.1.1.2.2 Type III printing press.- The installation and maintenance tools and operating supplies shall be placed in the appropriate cabinet. Cushioning material or blocking and bracing shall be used, as necessary, to prevent damage to the tools and supplies while in transit. The components of the printing press shall be secured in position on the platform base. Technical publications not preserved method IC-1 shall be placed in a paper envelope and secured in the appropriate cabinet. The fabric covers provided shall be installed over the motor generator, press and the packaged press, and secured.

5.1.2 Level C.- Presses shall be preserved and packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.- Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A.-

5.2.1.1 Type I and II printing presses.- Each type I or II printing press, preserved and packaged as specified in 5.1, shall be packed in a crate conforming to type I or II, class 1 or 2, style a or b of MIL-C-104. The motor generator, tools, and fiberboard boxes containing the plate bending jig and eyelet machine shall be secured in an unused portion of the crate. Contents shall be anchored in accordance with the appendix of MIL-C-104.

5.2.1.2 Type III printing press.- Unless otherwise specified, type III printing presses, preserved and packaged as specified in 5.1, shall require no overpacking. When specified (see 6.2), each type III printing press shall be packed in a crate conforming to type I or II, class 1 or 2, style a or b of MIL-C-104, and anchored in accordance with the appendix of MIL-C-104.

5.2.2 Level C.- Printing presses, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Additional marking.- Type III printing press, packed in a crate as specified in 5.2.1.2, shall have each side of the crate marked in letters minimum 1-inch high, as follows:

HEAVY LOAD

_____LBS.

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6. NOTES

6.1 Intended use.- The printing presses covered by this specification are for large volume, high speed production of special warfare propaganda leaflets in the field.

6.2 Ordering date.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type press required (see 1.2).
(c) Maintenance tools for type I, II and III presses required (see 3.6.2).
(d) Operating supplies required for type III press (see 3.7).
(e) Spare parts for type I, II and III presses required (see 3.9).
(f) Text of material to be printed and ink color (see 4.5.1).
(g) Selection of applicable levels of preservation and packaging, and packing see 5.1 and 5.2).
(h) When crates are required for type III presses for level A (see 5.2.1.2 and 6.4).

6.3 Contract data requirement.- Any requirements for equipment manuals for the item covered by this specification should be included in DD Form 1423, Contract Data Requirements List and cited in contract.

6.4 Crating.- Crates should be specified for type III printing presses for level A pack when it is anticipated that they will be subject to superimposed loads while in transit (see 6.2).

Custodian:

Army - GL

Preparing activity:

Army - GL

Project No. 3610-A170

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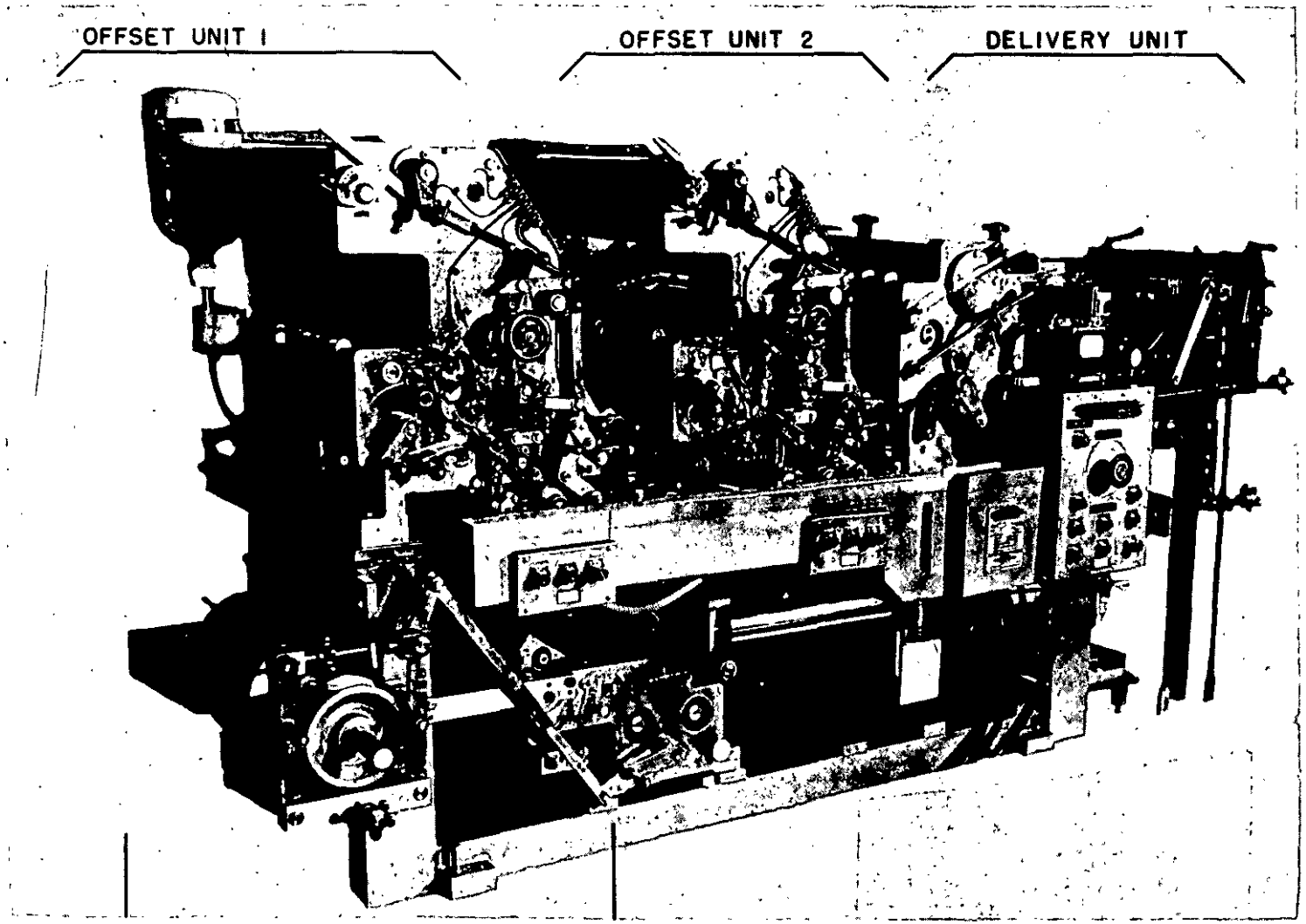


Figure 1. Type I - 2 Color Press (1 color one side and 1 color other side of the web or 2 colors one side of the web.)

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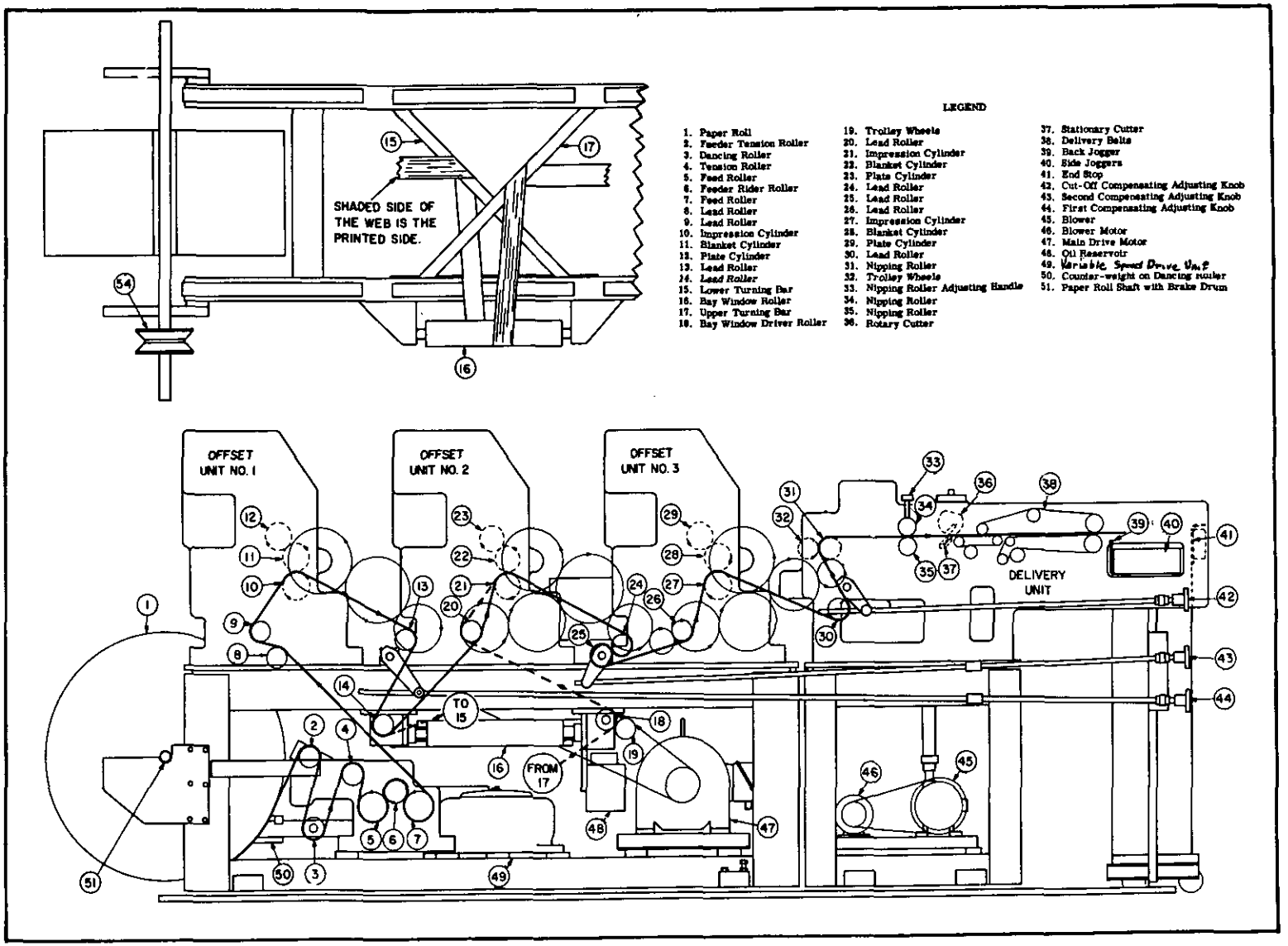
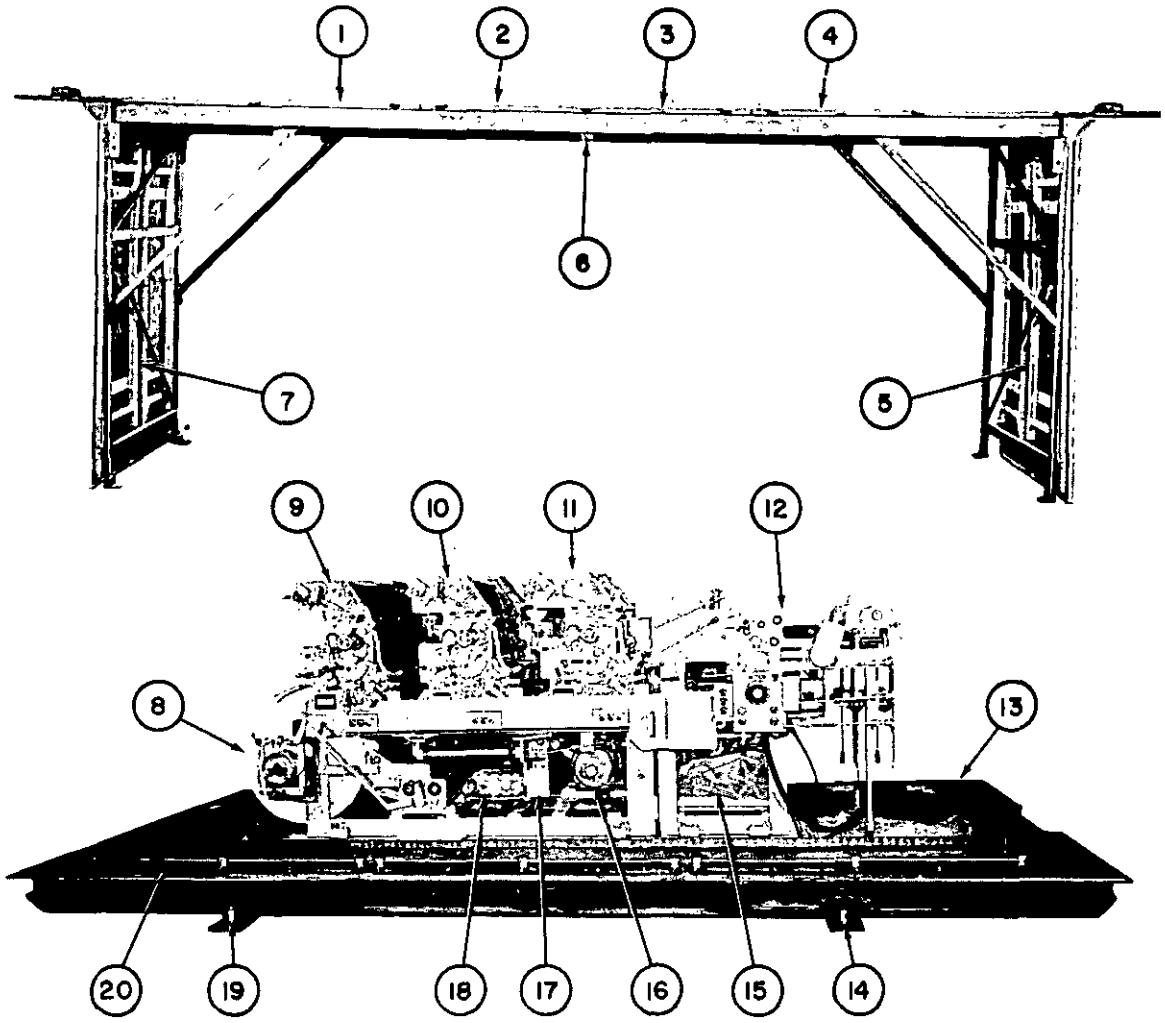


Figure 2. Type II-3 Color Press (1-color one side and 2-colors other side of the web or 3-colors one side of the web) and Method of Threading the Web of Web-Fed Lithographic Presses.

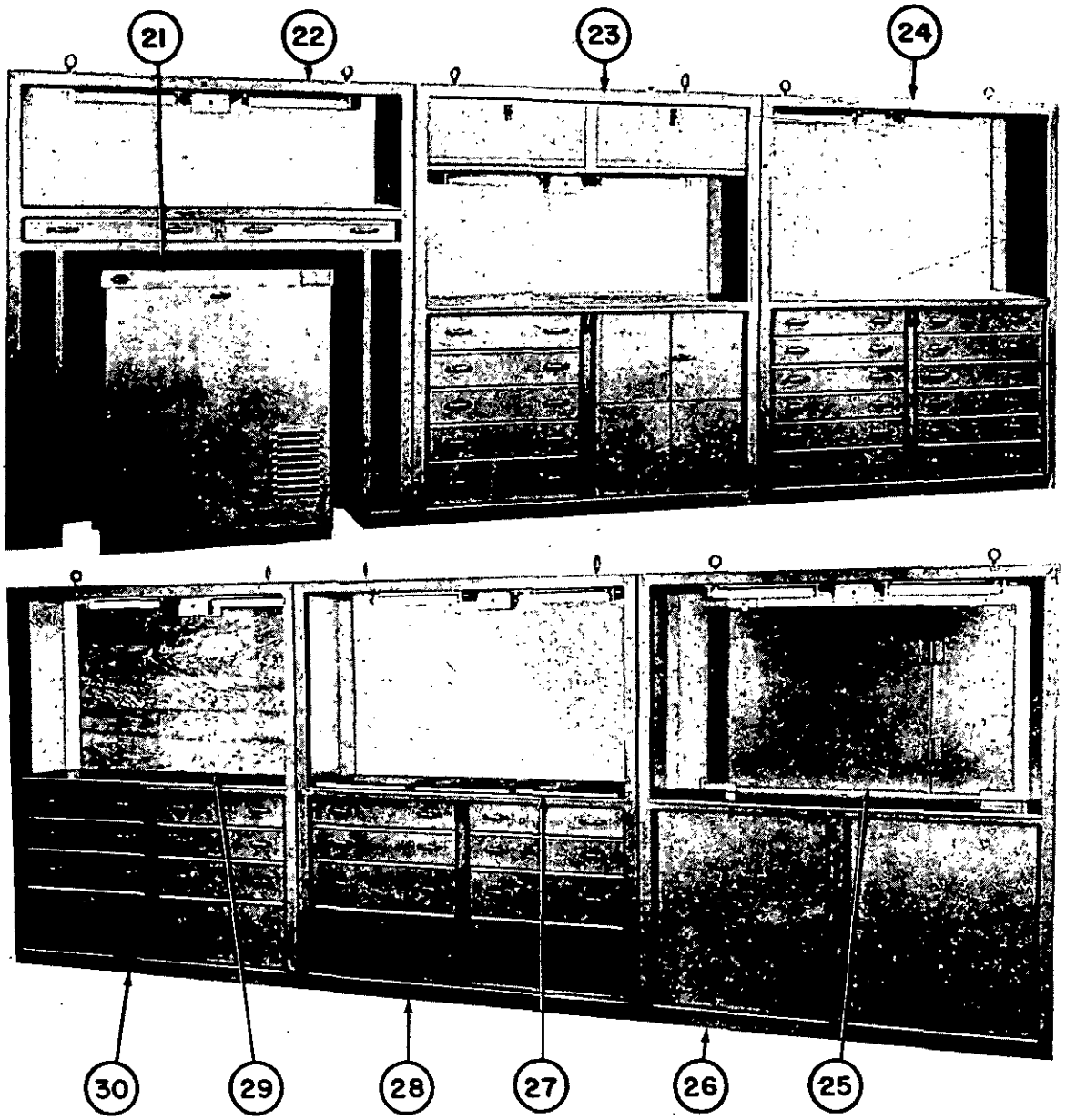
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1. Platform Extension Plate (With Legs)
2. Ramp Plate (For Removing Storage Cabinets)
3. Platform Extension Plate (With Legs)
4. Ramp Plate (For Removing Storage Cabinets)
5. Tables (two)
6. Shipping Frame (Knock-Down Type)
7. Tables (two)
8. Feed-Roll Stand and Brake Mechanism
9. Offset Unit No. 1
10. Offset Unit No. 2
11. Offset Unit No. 3
12. Delivery Unit
13. Platform Base
14. Leveling Jack
15. Blower and Motor Unit
16. Main Drive Motor
17. Oil Reservoir
18. Variable Speed Drive Unit
19. Leveling Jack
20. Brackets

Figure 3. Type III - 3 Color Press With U.S.Army Press Package

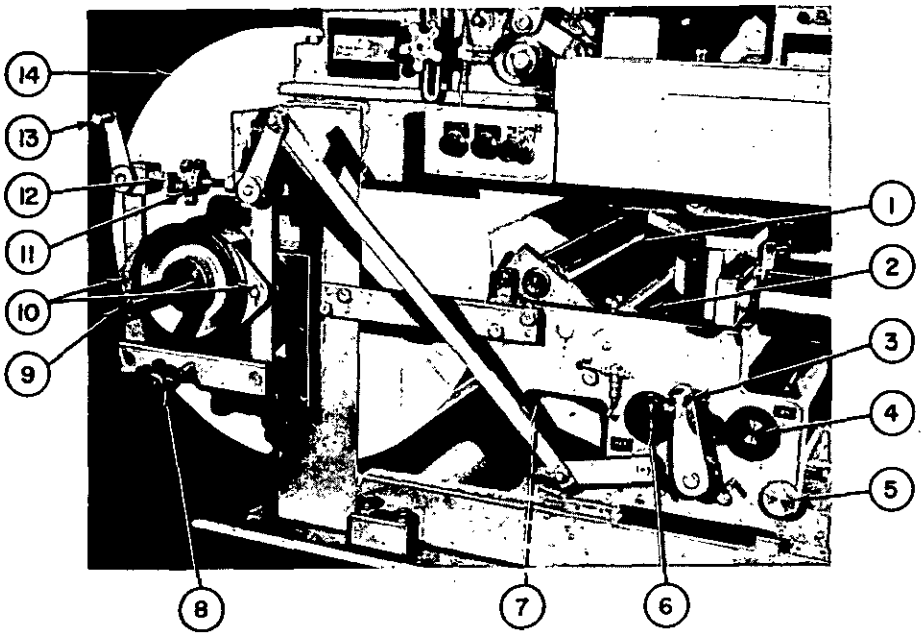
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21. Motor-Generator Unit
22. Storage Cabinet A
23. Storage Cabinet B
24. Storage Cabinet C
25. Platform Base Ramp
26. Storage Cabinet F
27. Ink Mixing Slabs (three)
28. Storage Cabinet E
29. Tool Board
30. Storage Cabinet D

Figure 3 (cont.) Type III -3 Color Press With U.S.Army Press Package

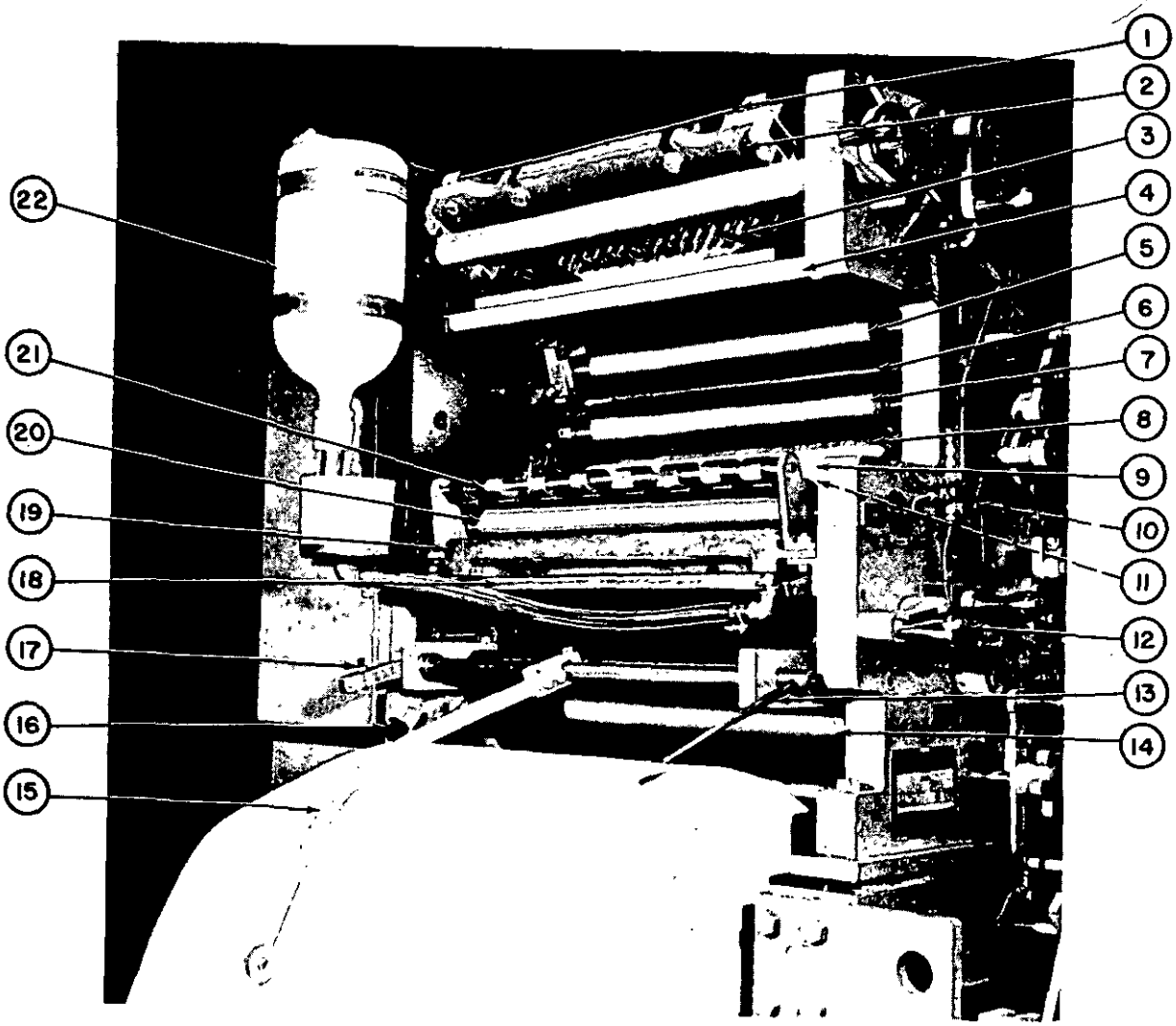
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1. Feeder Tension Roller
2. Tension Roller
3. ON-OFF Feeder Rider Roller Handle
4. Feeder Roller
5. Variable Speed Drive Control Knob
6. Feeder Roller
7. Dancing Roller
8. Paper Roll Lateral Adjusting Knob
9. Paper Roll Shaft with Brake Drum
10. Brake Shoes (two)
11. Brake Tension Adjusting Knob
12. Brake Locking Nut
13. Brake Locking Handle
14. Paper Roll

Figure 4. Feed-Roll Stand and Brake Mechanism

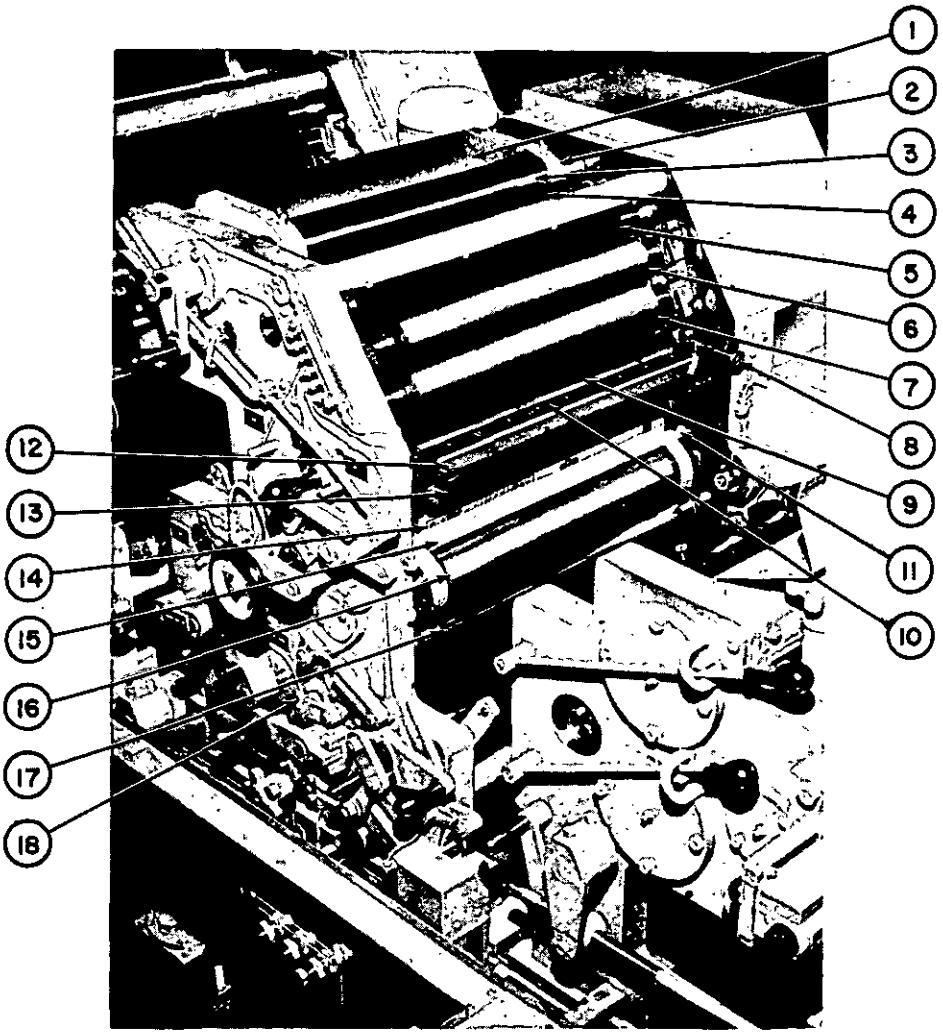
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1. Ink Fountain Blade
2. Ink Fountain
3. Ink Fountain Keys (twenty-two)
4. Ink Drip Pan
5. Rubber-Covered Distributor Roller (two)
6. Ebonite Vibrator Roller
7. Rubber-Covered Form Roller
8. Plate Cylinder
9. Water Form Roller (two)
10. Water Vibrator Roller
11. Water Ductor Roller
12. Offset Unit Operating Controls - See Figure 8
13. Paper Roll Limit Switch Handle
14. Lead Roller
15. Paper Roll Limit Switch Arm
16. Paper Roll Limit Switch
17. Paper Roll Limit Switch Adjustment Screw
18. Water Drip Pan
19. Water Fountain
20. Water Fountain Roller
21. Water Fountain Stops (seven)
22. Water Solution Container and Holder

Figure 5. Offset Unit, Rear End

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1. Ink Fountain Blade
2. Ink Fountain
3. Ink Fountain Roller
4. Ink Ductor Roller
5. Ebonite Vibrator Roller
6. Rubber-Covered Distributor Roller
7. Rubber-Covered Distributor Roller
8. Wash-Up Adjusting Screw (two)
9. Ebonite Vibrator Roller
10. Wash-Up Blade
11. Cylinder Guard Fastening Screw
12. Wash-Up Attachment
13. Rubber-Covered Form Roller
14. Plate Cylinder
15. Cylinder Guard
16. Blanket Cylinder
17. Impression Cylinder
18. Offset Unit Operating Controls - See Figure 8

Figure 6. Offset Unit, Front End

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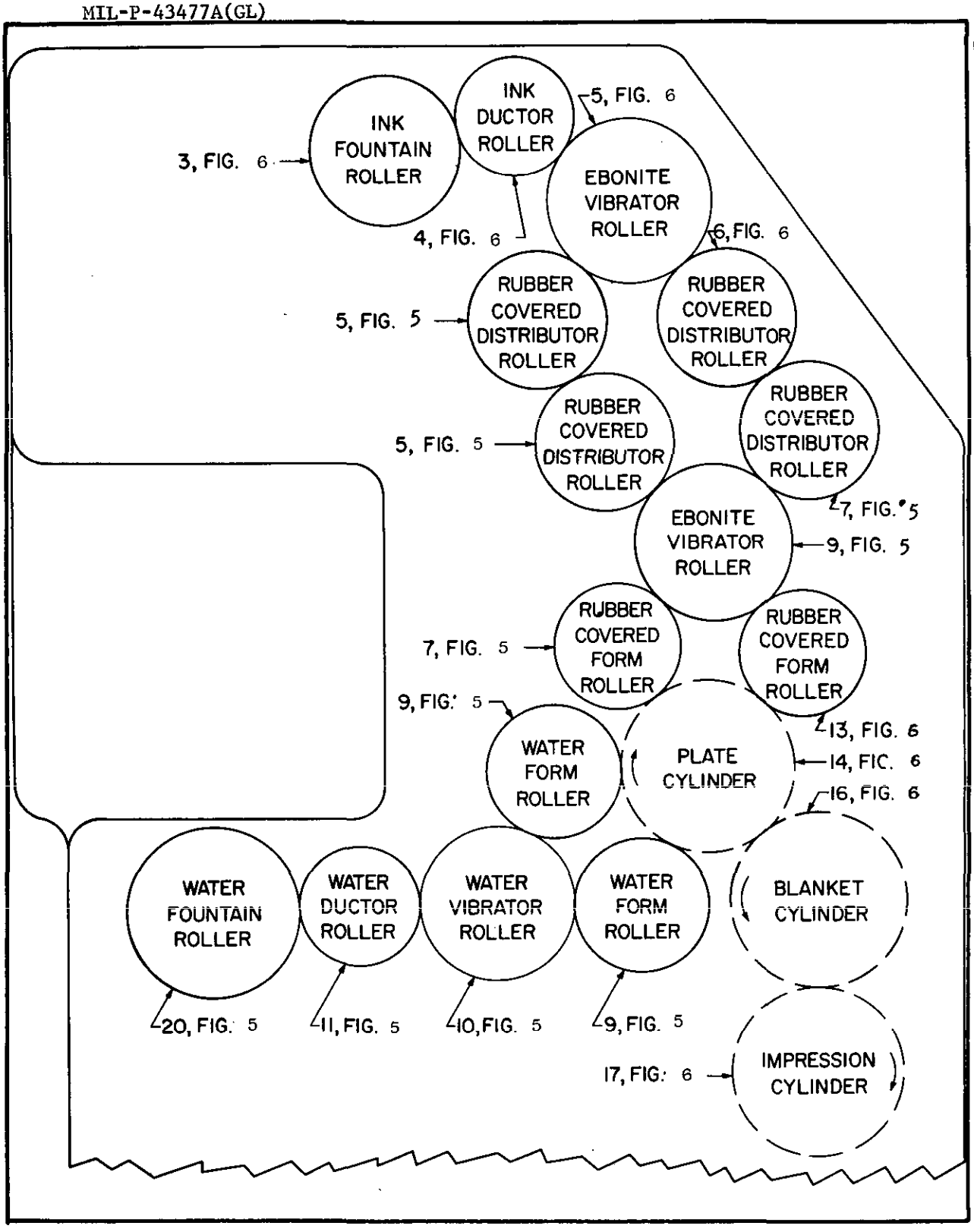
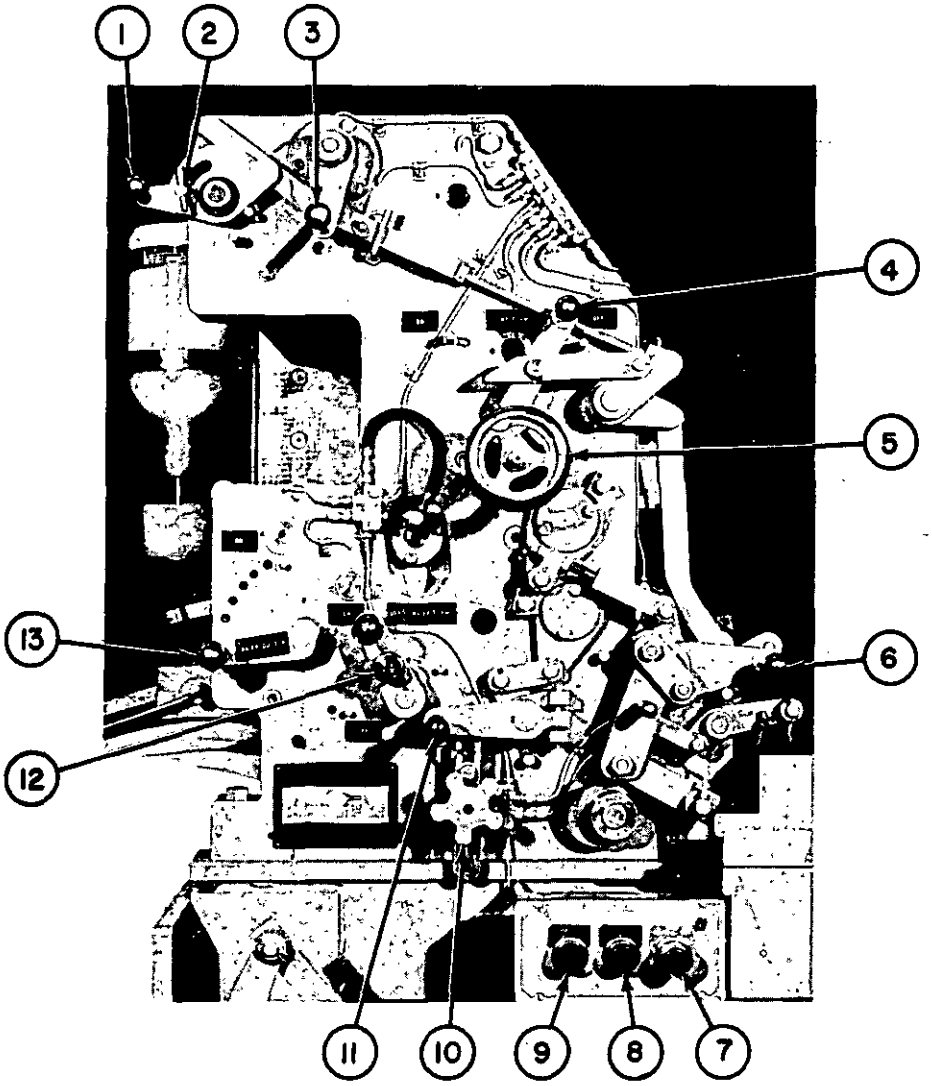


Figure 7. Arrangement of Rollers and Cylinders of Offset Unit

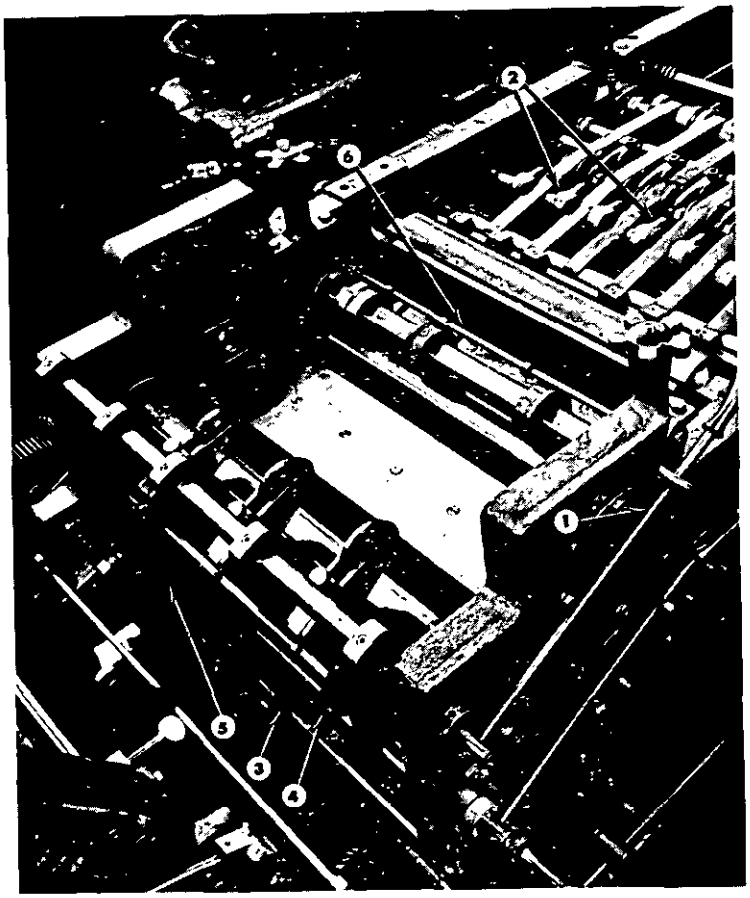
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1. Ink Fountain Handle
2. Ink Ratchet Setting Screw
3. Ink Fountain Roller Handle and Pawl
4. INK FORM, ON-OFF Handle
5. Plate Cylinder Lateral Control Wheel
6. Impression Handle (With Solenoid Switch)
7. ON-STOP-SAFE Switch
8. JOG REV Switch
9. JOG FOW Switch
10. Impression Cylinder Adjusting Knob
11. Impression Cylinder Throw-Off Handle
12. WATER FORM, ON-OFF Handle
13. WATER DUCTOR, ON-OFF Handle

Figure 8. Offset Printing Unit Operating Controls

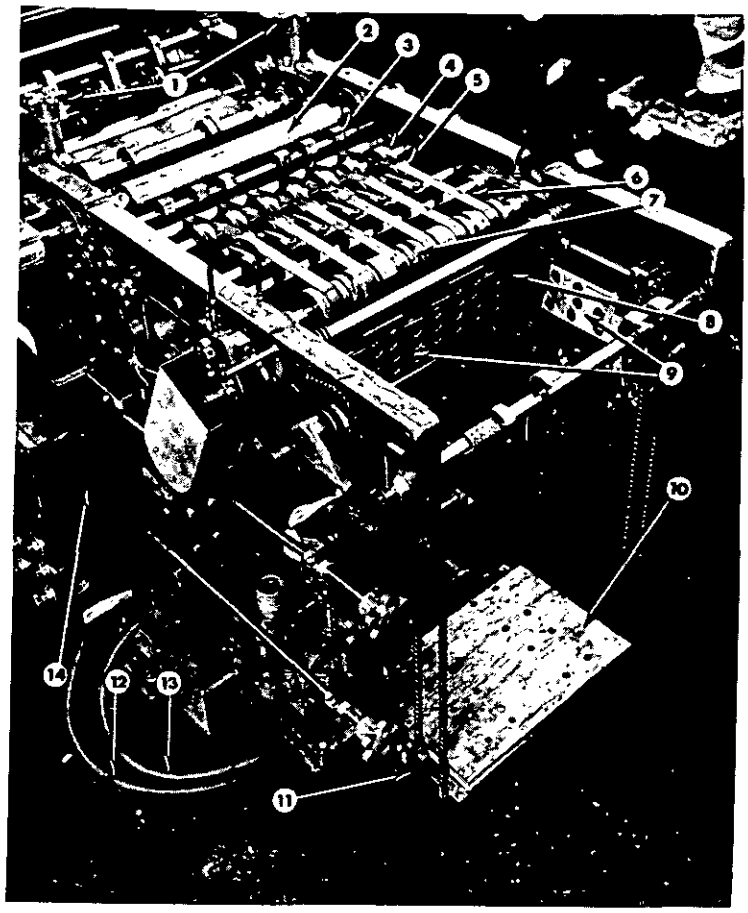
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1. Delivery Unit Operating Controls See Figure 11
2. Sheet Kickers (seven)
3. Lead Roller
4. Trolley Wheel
5. Trolley Wheel
6. Mechanical Trip Mechanism.

Figure 9. Delivery Unit, Rear End

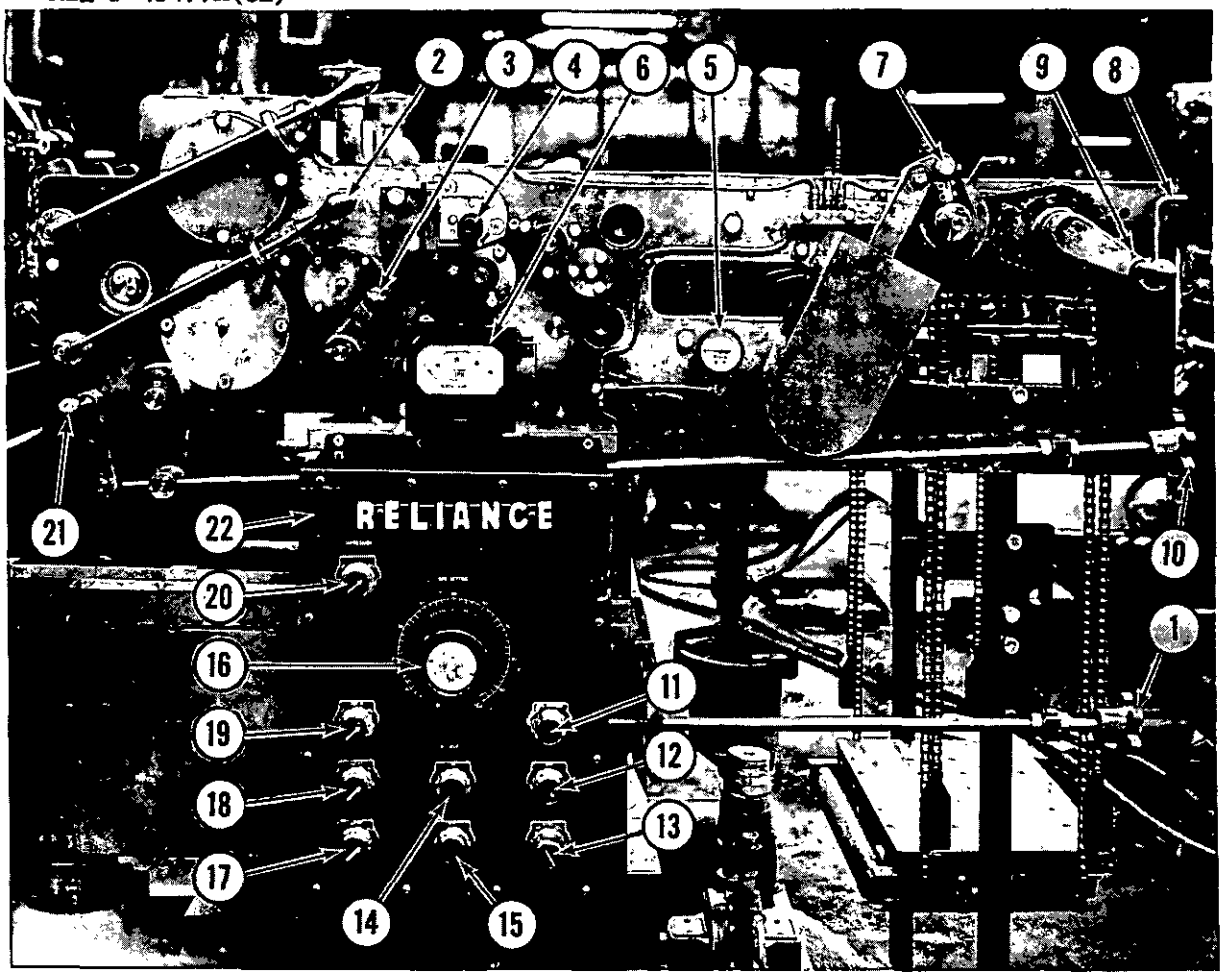
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1. Nipping Roller and Adjustment Knobs (two)
2. Rotary Cutter
3. Paper Guides (six)
4. Sheet Kickers (seven).
5. Delivery Belt Adjusting Pulleys
6. Delivery Belt
7. Knurled Delivery Roller
8. Static Eliminator Bar
9. Side and End Joggers
10. Delivery Table With Castors
11. Chain and Chain Weights
12 Press Control Cable
13. Static Eliminator Cable
14 Delivery Unit Operating Controls See Figure 11

Figure 10. Delivery Unit, Front End

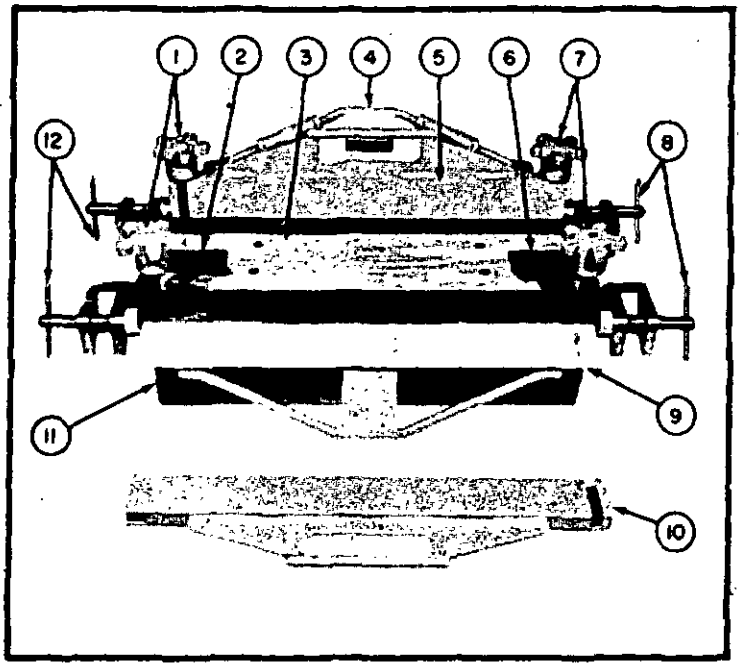
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1. Register Control Knob
2. Trolley Wheels Handle
3. Nipping Roller Adjusting Handle
4. Sheet Counter
5. Oil Pressure Gage
6. Speed Indicator
7. Automatic Sheet Lowering
8. Sheet Stop Handle
9. Manual Sheet Lowering and Raising Handle
10. Cut-Off Compensating Adjusting Knob
11. Jog Forward Switch
12. Run Switch
13. On-Stop-Safe Switch
14. Blower, On Switch
15. Blower, Off Switch
16. Speed Setting Control
17. On-Off Impression Switch No. 3
18. On-Off Impression Switch No. 2
19. On-Off Impression Switch No. 1 (Not Used on 2-Color Press)
20. On-Off Master Impression Switch
21. Delivery Clutch Handle
22. Control Station

Figure 11.-Delivery Unit and Main Control Station Operating Controls.

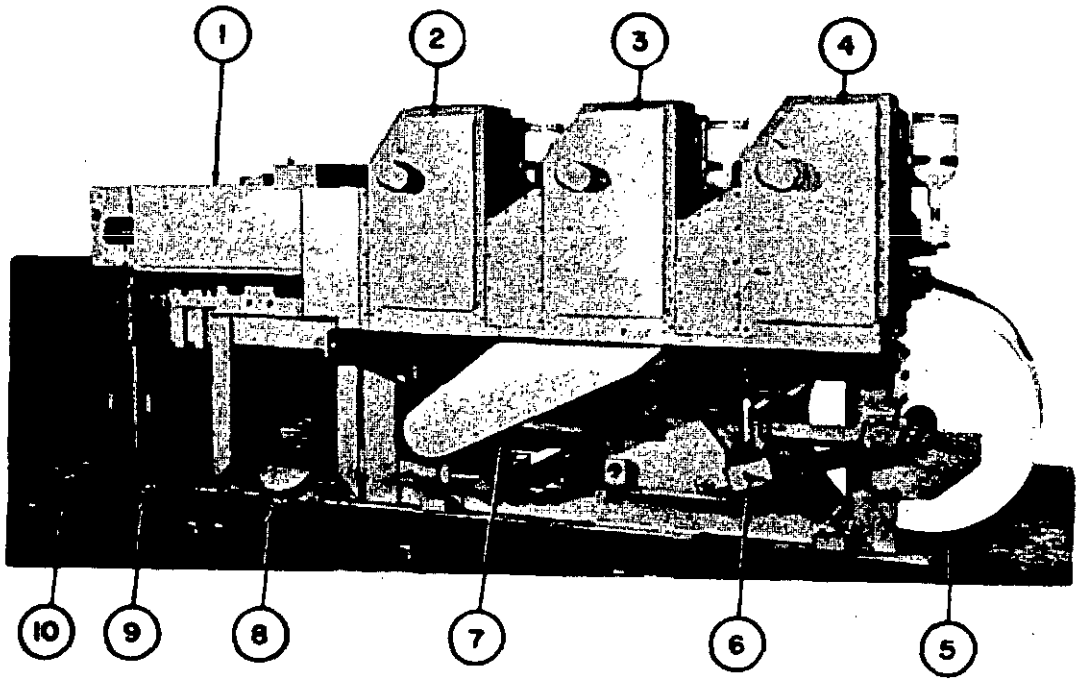
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1. Hand Knobs (two)
2. Guide Plate
3. Plate Bending Jig Board
4. Tail Edge Bending Bar
5. Tail Edge Stationary Bar
6. Guide Plate
7. Hand Knobs (two)
8. Bending Bar Pins (two)
9. Gripper Edge Bending Bar
10. Gripper Edge Stationary Bar
11. Plate Bending Jig Frame
12. Bending Bar Pins (two)

Figure 12. Plate Bending Jig

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1. Safety Guard Door for Delivery Unit
2. Safety Guard Door for Offset Unit No. 3
3. Safety Guard Door for Offset Unit No. 2
4. Safety Guard Door for Offset Unit No. 1
5. Paper Roll
6. Counterweight for Dancing Roller
7. Safety Guard for Main Drive Wheels and V-Belt
8. Press Control Cable
9. Static Eliminator Cable
10. Blower Cable

Figure 13 Safety Guards and Rear of Offset Units and Delivery Unit

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Printing Press, Offset, Web-Fed, For Special Warfare MIL-P-43477A(GL)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity).*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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